



Department of Artificial Intelligence and Machine Learning

Date: 7.11.2025	Test -1	Max. Marks: 50+10
Semester: V	UG	Duration: $1\frac{1}{2}$ Hrs + $\frac{1}{2}$ Hr
Course Title: Artificial Neural Networks and Deep Learning		Course Code: AI253IA

Part A

Sl No	Questions	M	BT	CO
1.	A perceptron has weights $w_1 = 0.4$, $w_2 = \text{and } -0.2$, $b = 0.1$. For input $x_1 = 1$, $x_2 = 0$, compute the net input and output (step activation).	1	3	1
2.	Compute the output of a sigmoid neuron if $\text{net} = 0.8$.	1	2	1
3.	Define learning rate. Why is it important in neural network training?	1	2	2
4.	Given inputs $x_1 = 1$, $x_2 = 2$, desired output $d = 8$, initial weights $w = [0.5, 1.0]$, and $\eta = 0.1$. Compute updated weights using the LMS rule after one iteration.	2	4	2
5.	Mention two advantages of using multilayer perceptrons over single-layer perceptrons.	2	3	1
6.	Compute the total number of weights in a 3-2-1 MLP (3 inputs, 2 hidden, 1 output; include bias).	2	3	1
7.	What are the sequential and batch modes of training?	1	2	2

Part B

Sl No	Questions	M	BT	CO
1a.	Explain the structure and mathematical model of an artificial neuron. Discuss how the activation function affects the neuron's decision boundary.	04	2	1
1b.	A neuron receives three inputs: $x_1 = 1$, $x_2 = 0$, $x_3 = 1$ with weights $w_1 = 0.2$, $w_2 = -0.3$, $w_3 = 0.4$, and bias $b = 0.1$. i. Compute the net input and the output using a sigmoid activation. ii. Recompute output if $b = 0.5$ and comment on its effect.	06	3	1
2a.	Compare and contrast Hebbian, Competitive, and Error-Correction learning. Give one practical example where each learning rule can be applied.	05	3	2
2b.	Given an adaptive filter with $x = [1, 2]$, $w = [0.5, -0.6]$, $d = 3$, and $\eta = 0.05$, compute: 1. Output $y = w^T x$ 2. Error $e = d - y$ 3. Updated weights $w_{\text{new}} = w + \eta ex$.	05	4	2
3	Explain the Perceptron Learning Algorithm with its update rule and convergence theorem. Why does the perceptron fail for non-linearly separable problems?	06	2	2
4	The network weights have been initialized as shown in Fig.1. Analyze the Multilayer perceptron Back Propagation algorithm for the initialized network by doing the following (use the sigmoid function): with a learning rate $\eta = 0.25$. i. Compute the net input and output of each hidden neuron and the final output neuron using the forward pass.	14	4	3

- ii. Calculate the error term, the delta values for both output and hidden neurons, and determine the gradient direction.
- iii. Update the weights using the backpropagation learning rule and show how the error decreases over two iterations of training.

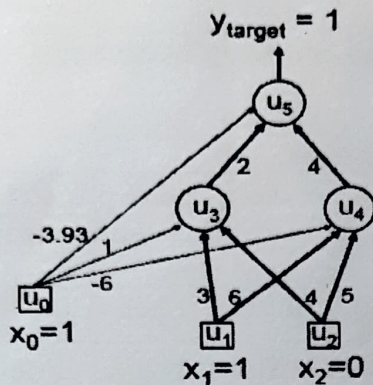


Fig. 1.

- 5 Consider the single-neuron perceptron network shown in Figure 2 below. The network is used to classify vehicles as either a BUS (+1) or a CAR (0) based on three input parameters: x_1 : Weight, x_2 : Fuel Capacity, x_3 : Number of Passengers. A bias input $x_0 = 1$ is also provided.

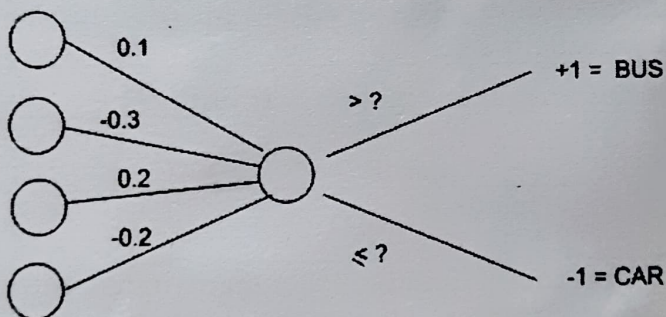


Fig. 2.

Given: Learning rate, $\eta = 0.1$, Activation: Sigmoid, Input vector (with bias): $x = (1, 10, 12, 13)$, Target output: $t = 1$ (BUS)

Tasks

- i. Compute the net input and the sigmoid output $y = f(\text{net})$ for the given input. State whether the perceptron currently classifies it as a car or a bus.
- ii. Suppose this input was misclassified. Using the sigmoid delta learning rule, compute the weight corrections:

$$\Delta w_i = \eta(t - y)y(1 - y)x_i \quad \text{for } i = 0, 1, 2, 3$$
- iii. Determine the new weights after this update.
- iv. Using the updated weights, calculate the new output y' for the same input. Does the network now correctly classify the example?
- v. A second input $(1, 5, 7, 7)$ was previously correctly classified as a car. After weight updates, discuss briefly whether this example may now be misclassified (over-correction).

Threshold = 0.5



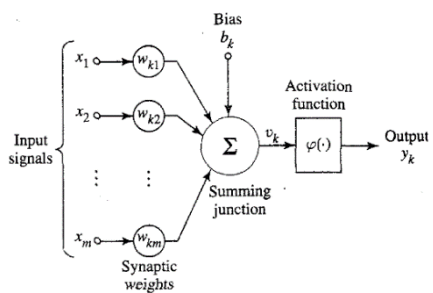
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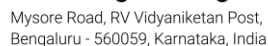
	Test -1 –Scheme and Solution	Max. Marks: 50+10
Semester: V	UG	Duration: 1 $\frac{1}{2}$ Hrs + $\frac{1}{2}$ Hr
Course Title: Artificial Neural Networks and Deep Learning		Course Code: AI253IA

Part A

SI No	Questions	M	BT	CO
1.	A perceptron has weights $w_1 = 0.4$, $w_2 = \text{and } -0.2$, $b = 0.1$. For input $x_1 = 1$, $x_2 = 0$, compute the net input and output (step activation). 0.5 and 1	1	3	1
2.	Compute the output of a sigmoid neuron if $net = 0.8$. 0.6900	1	2	1
3.	The learning rate (η) is a positive scalar that scales weight updates during training (how big a step we take in parameter space). If η is too large training may diverge or oscillate; if too small convergence is very	1	2	2
4.	0.5, 1.05, 2.10	2	4	2
5.	Can learn nonlinearly separable functions (hidden layers let them combine features to solve problems single-layer perceptrons cannot, e.g., XOR). Can approximate complex mappings (universal approximation for continuous functions) and learn hierarchical features.	2	3	1
6.	1 1	2	3	1
7.	Sequential (online): update weights after each training example — faster immediate updates, introduces noise that can help escape local minima. Batch: accumulate errors/gradients over the whole training set and update weights once per epoch — smoother, more stable gradient estimate but needs more memory and slower per-epoch updates.	1	2	2

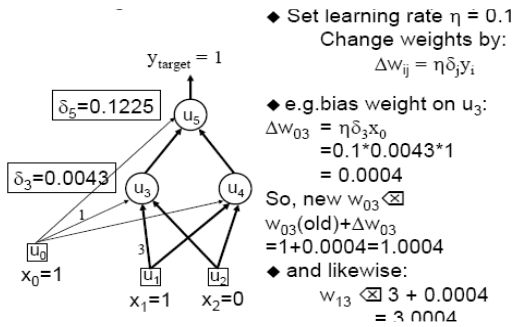
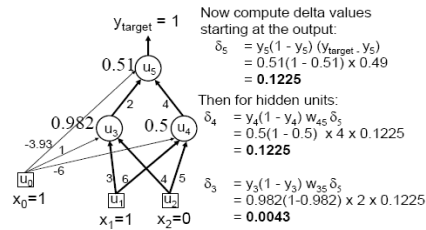
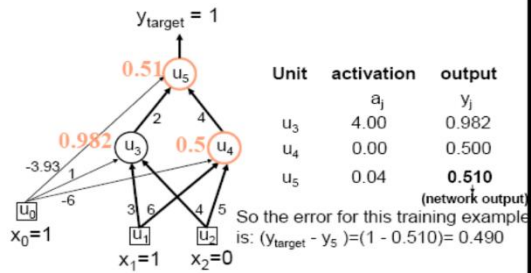
Part B

SI No	Questions	M	BT	CO
1a	 - 2M, Explanation -2M	04	2	1

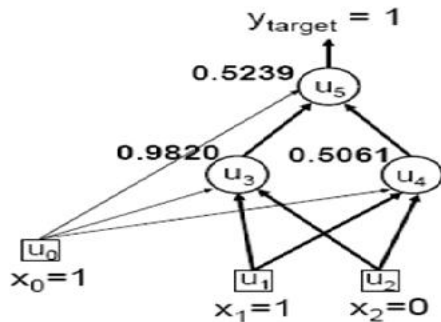


1b	Bias (b)	Net Input	Sigmoid Output	Observation			0 6	3	1
	0.1	0.7	0.668	Normal activation -3M					
	0.5	1.1	0.750	Higher bias → higher output-3M					
2a	Learning Rule	Basic Principle	Weight Update Formula (Conceptual)	Type of Learning	Key Features	Practical Example	0 5	3	2
	Hebbian Learning	“Neurons that fire together, wire together.” — strengthens connections between simultaneously active neurons.	$\Delta w = \eta \cdot x \cdot y$ (where x = input, y = output)	Unsupervised	Reinforces correlations between inputs and outputs; no error or teacher signal.	Associative memory (e.g., pattern storage in Hopfield networks).			
	Competitive Learning	Neurons compete; only one (winner) neuron updates its weights toward the input (Winner-Takes-All).	Only winning neuron’s weights updated: $\Delta w = \eta \cdot (x - w)$	Unsupervised	Promotes specialization — neurons become tuned to different input patterns or clusters.	Clustering / Self-Organizing Maps (Kohonen networks)			
	Error-Correction Learning	Adjusts weights to minimize the difference (error) between desired and actual outputs.	$\Delta w = \eta \cdot (d - y) \cdot x$	Supervised	Uses a teacher signal; aims to reduce prediction error; basis for backpropagation.	Perceptron / MLP training for classification or regression			
2b	Quantity	Expression	Value				0 5	4	2
	Output (y)	$(w^T x)$	−0.7						
	Error (e)	$(d - y)$	3.7						
	Updated Weights	$(w + \eta e x)$	[0.685, −0.23]						
3	Explain the Perceptron Learning Algorithm with its update rule and convergence theorem. Why does the perceptron fail for non-linearly separable problems?								

4



i	j	w_{ij}	δ_j	y_i	Updated w_{ij}
0	3	1	0.0043	1.0	1.0004
1	3	3	0.0043	1.0	3.0004
2	3	4	0.0043	0.0	4.0000
0	4	-6	0.1225	1.0	-5.9878
1	4	6	0.1225	1.0	6.0123
2	4	5	0.1225	0.0	5.0000
0	5	-3.92	0.1225	1.0	-3.9078
3	5	2	0.1225	0.9820	2.0120
4	5	4	0.1225	0.5	4.0061



On next forward pass:
The new activations are:
 $y_3 = f(4.0008) = 0.9820$
 $y_4 = f(0.0245) = 0.5061$
 $y_5 = f(0.0955) = 0.5239$
Thus the new error
 $(y_{\text{target}} - y_5) = (1 - 0.5239) = 0.476$
has been reduced by 0.014
(from 0.490 to 0.476)

1
4

4

3

5

5) a)
$$a) = (0.1 \times 1) + (-0.3 \times 10) + (0.2 \times 12) + (-0.2 \times 13)$$

$$= 0.1 + (-3) + (2.4) + (-2.6)$$

$$= -3.1$$

net input.

sigmoid output $y = f(\text{net})$

$$y = f(-3.1)$$

Activation function used is sigmoid

$$y = \frac{1}{1 + e^{-x}} = \frac{1}{1 + e^{-(-3.1)}} = 0.0431$$

$y < 0.5 \Rightarrow$ misclassified. (not bus / car)

b) $(1-y) = 1 - 0.0431 = 0.9569$
 $y(1-y) = 0.0431 \times (1 - 0.0431) = 0.0413$

$$y(1-y) = 0.1 \times 0.9569 \times 0.0413 = 0.00394$$

$$\Delta W_i = y(1-y) \cdot x_i$$

Considering $y(1-y) = K$

$$\Delta W_0 = K \cdot x_0 = 0.00394 (1) = 0.00395$$

$$\Delta W_1 = K \cdot x_1 = 0.00394 (10) = 0.03947$$

$$\Delta W_2 = K \cdot x_2 = 0.00394 (12) = 0.04737$$

$$\Delta W_3 = K \cdot x_3 = 0.00394 (13) = 0.05131$$

1
0

4

2



	iii) Determine the new weights after the update: $W_i = W_i + \Delta W_i$ $W_0 = 0.1 + 0.00395 = 0.10395 \Rightarrow \approx 0.104$ $W_1 = -0.3 + 0.03947 = -0.260 \approx -0.261$ $W_2 = 0.2 + 0.04737 = 0.247 \approx 0.247$ $W_3 = -0.2 + 0.05131 = -0.14869 \approx -0.149$ iv) net = $W_0 x_0 + W_1 x_1 + W_2 x_2 + W_3 x_3$ $= 0.1039(1) + (-0.261 \times 10) + (0.247 \times 12) + (-0.15 \times 13)$ $= 0.1039 - 2.61 + 2.964 - 1.95$ $= -1.4921$ $y = \frac{1}{1 + e^{-1.49}} = 0.188 \approx 0.20 \quad \text{again } y < 0.5$ misclassified - so it is 'Car'. v) $(1, 5, 7, 7) = -1.4$ $\sigma(-1.4) = 0.1981 \quad \text{Correctly classified as Car.}$ so for new weights, $\text{net}_{\text{new}} = 0.103(1) + (-0.260)(5) + (0.247(7)) + (-0.14)(7) = -0.508$ $\sigma(-0.508) = 0.376 < 0.5$ $\therefore$ output has increased from 0.198 to 0.376.				
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M-Marks, BT-Blooms Taxonomy Levels, CO-Course Outcomes

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks CIE	16	30	14	-	-	-	13	16	31	-	-